7. Write a program to Implementing Advanced Regularization Techniques and Model Enhancements.

```
import numpy as np
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout,
BatchNormalization
from tensorflow.keras.regularizers import l1, l2
from tensorflow.keras.callbacks import EarlyStopping

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature Scaling (Enhancement)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Build Neural Network Model
model = Sequential()

# Hidden Layer 1 with L2 regularization + BatchNorm + Dropout
model.add(Dense(64, activation='relu', kernel_regularizer=l2(0.01),
input_shape=(X.shape[1],)))
model.add(BatchNormalization())
model.add(Dropout(0.5))

# Hidden Layer 2 with L1 regularization
model.add(Dense(32, activation='relu', kernel_regularizer=l1(0.01)))
model.add(Dropout(0.3))

# Output Layer
model.add(Dense(1, activation='sigmoid'))

# Compile model
model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
```

```
# Early Stopping (Enhancement)
early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True)

# Train model
history = model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=50,
    batch_size=16,
    callbacks=[early_stop],
    verbose=1
)

# Evaluate
loss, accuracy = model.evaluate(X_test, y_test)
print("Test Accuracy:", accuracy)

# Predictions
y_pred = (model.predict(X_test) > 0.5).astype(int)
print("Final Accuracy:", accuracy_score(y_test, y_pred))
```